

Algebra 1
Unit 11
Lesson 1 Practice

Name _____
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ growth factor and b represents ☐ initial amount

the ☐ growth factor.
☐ initial amount.

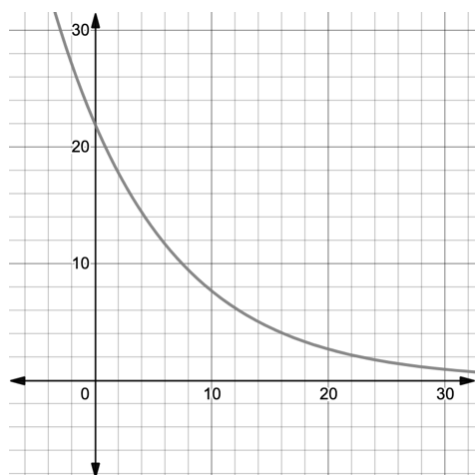
2. When b is between 0 and 1, the equation models ☐ exponential growth.
☐ exponential decay.

3. When b is greater than 1, the equation models ☐ exponential growth.
☐ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4. $y = 142(1.05)^x$ represents _____ because...

5. _____ represents _____
because...

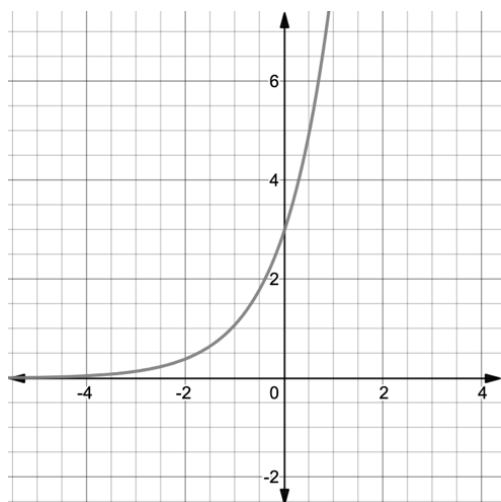


6. _____ represents _____ because...

x	y
-1	$\frac{1}{5}$
0	$\frac{1}{2}$
1	$\frac{5}{4}$
2	$\frac{25}{8}$

7. $y = 3(1 - 0.14)^x$ represents _____ because...

8. _____ represents _____ because...



9. Write an equation in $y = a(b)^x$ form that represents exponential growth.

10. Write an equation in $y = a(b)^x$ form that represents exponential decay.

Algebra 1
Unit 11
Lesson 2 Practice

Name _____
Date _____
Period _____

1. The graph of $y = -1(1.7)^x$ is

- ☐ increasing
☐ decreasing

and is

- ☐ positive.
☐ negative.

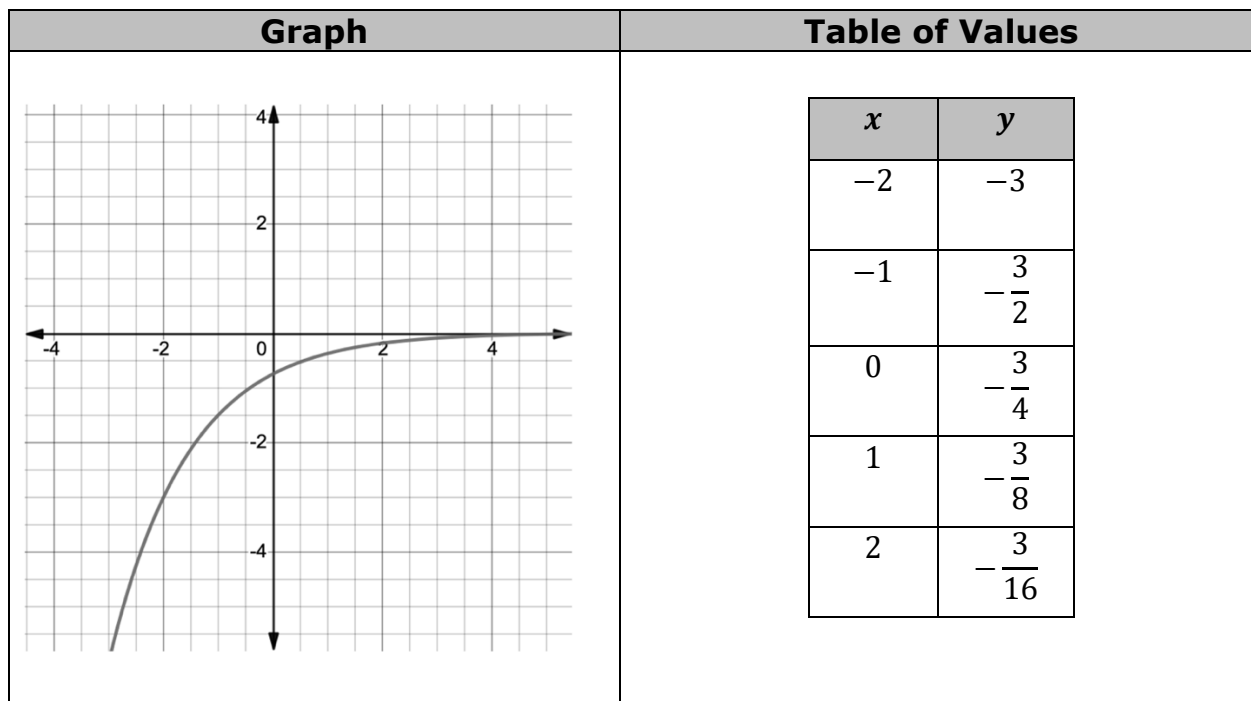
2. The graph of $y = \left(\frac{7}{3}\right)^x$ is

- ☐ increasing
☐ decreasing

and is

- ☐ positive.
☐ negative.

Use the graph and table of $y = -\frac{3}{4}(1 - 0.5)^x$ to determine key features.



Key Features			
3. Domain		4. Range	
5. y - intercept		6. Constant Percent Rate of Change	
7. Increasing or Decreasing		8. Positive	
		Negative	
9. Left End Behavior		10. Right End Behavior	

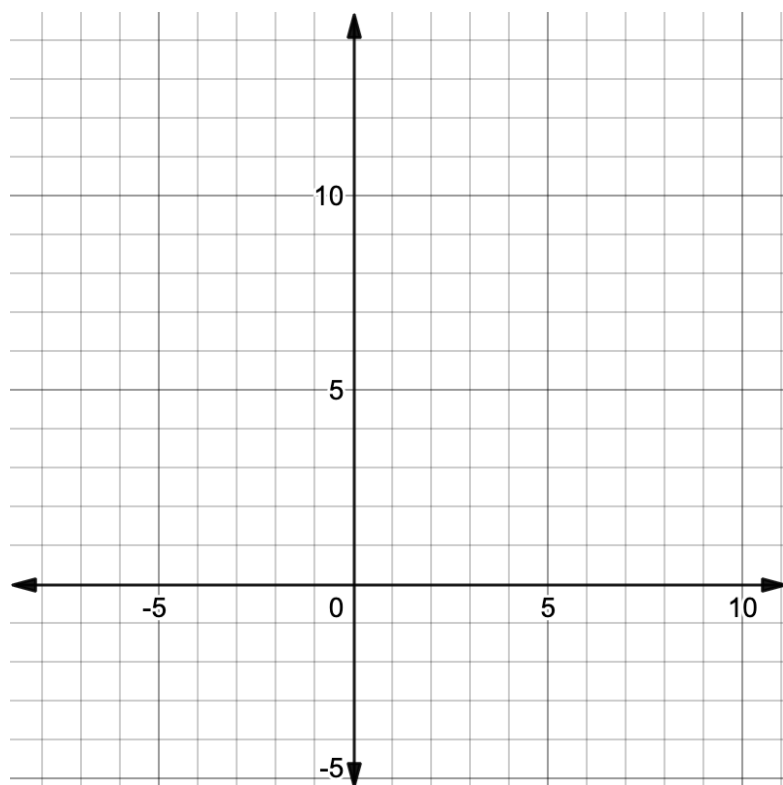
Algebra 1
Unit 11
Lesson 3 Practice

Name _____
Date _____
Period _____

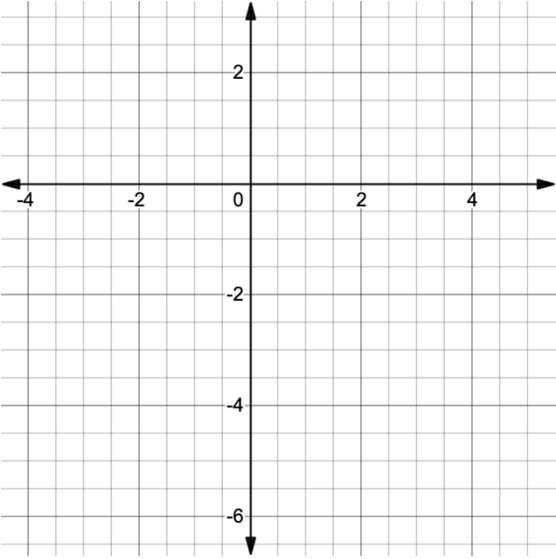
For questions 1-2, consider the function $f(x) = \frac{7}{2}(1.23)^x$.

1. Since a is ☐ positive
☐ negative and b is ☐ between 0 and 1,
☐ greater than 1, the function
- is ☐ increasing.
☐ decreasing. The left end behavior of the graph is ☐ $y \rightarrow 0$
☐ $y \rightarrow \infty$ and the
- right end behavior is ☐ $y \rightarrow 0$.
☐ $y \rightarrow \infty$. The y -intercept of the function is ☐ 0.23.
☐ 1.23.
☐ 3.5.

2. Sketch the graph of $f(x) = \frac{7}{2}(1.23)^x$.



Graph the function of $y = -(1 - 0.75)^x$, complete the table of values, and then determine all key features.

3. Graph	4. Table of Values												
	<table border="1"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-2</td><td></td></tr><tr><td>-1</td><td></td></tr><tr><td>0</td><td></td></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr></tbody></table>	x	y	-2		-1		0		1		2	
x	y												
-2													
-1													
0													
1													
2													

Key Features	
5. Growth or Decay	
6. Domain and Range	
7. Increasing/Decreasing	
8. Constant Percent Rate of Change	
9. Left and Right End Behavior	
10. Positive or Negative	

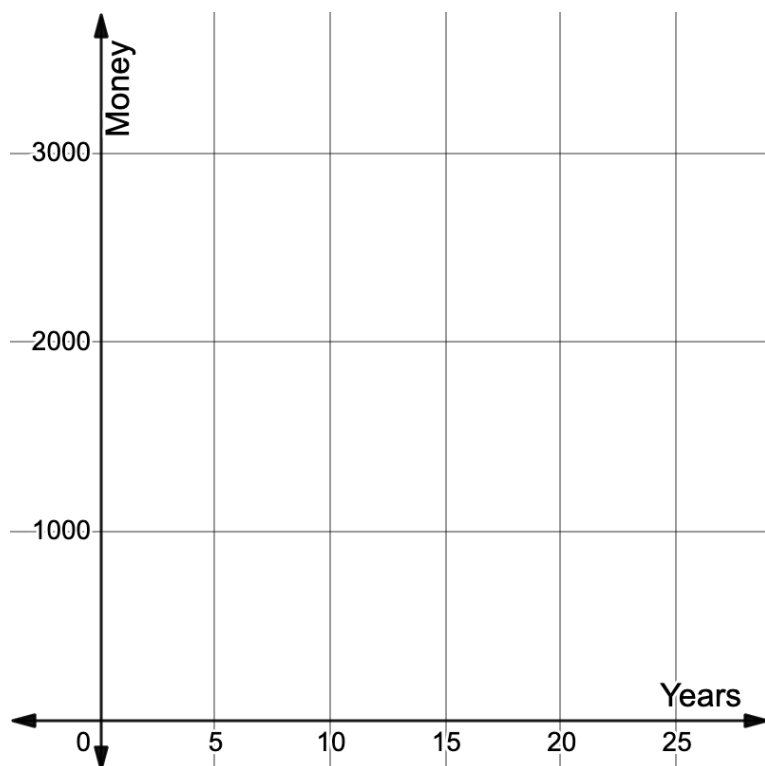
Algebra 1
Unit 11
Lesson 4 Practice

Name _____
Date _____
Period _____

Use the scenario below for questions 1-5.

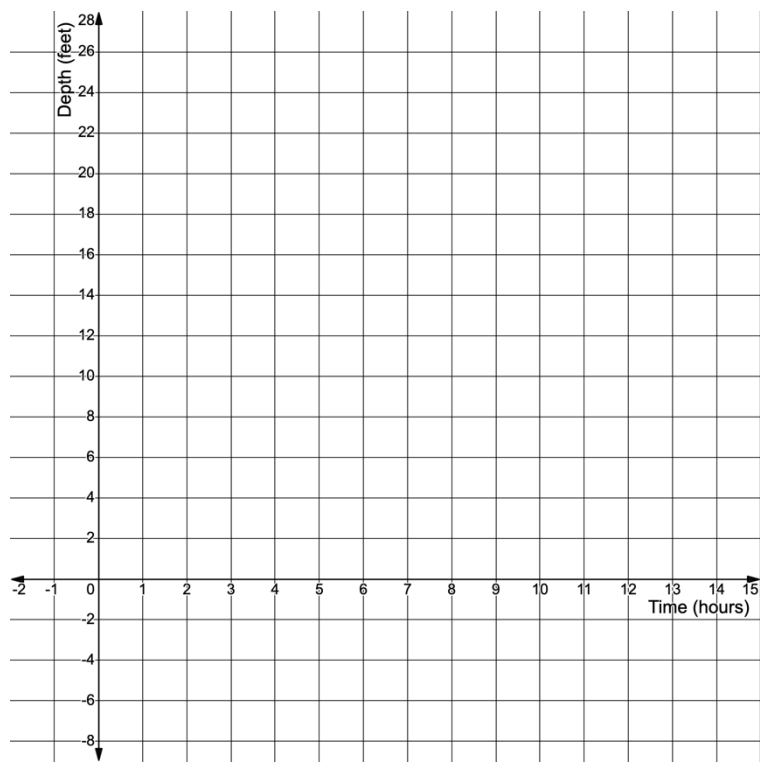
Janice started a bank account for her daughter with \$625, which earns 4.75% interest per year. The amount in her daughters account after x years can be modeled by the function $V(x) = \$625(1.0475)^x$.

1. Does the equation $V(x) = \$625(1.0475)^x$ represent exponential growth or decay?
2. What is the y -intercept and what does it represent in this situation?
3. Explain why the value of b is not 4.75, the interest rate.
4. Graph the equation $V(x) = \$625(1.0475)^x$.



The equation $y = 14\left(\frac{4}{5}\right)^x$ represents the depth of the water as a pool drains in feet in x hours. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6.	Domain	
7.	Range	
8.	y-intercept	
9.	Constant Percent Rate of Change	
10.	End Behavior	

Algebra 1
Unit 11
Lesson 5 Practice

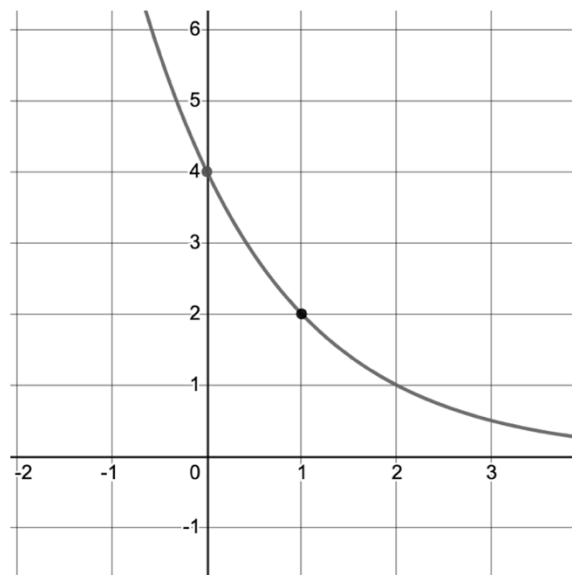
Name _____
Date _____
Period _____

The graph of an exponential function is shown.

1. Find the value of a by substituting the point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{} = a(b)^{\boxed{}}$$

2. Use the value of a found in question 1 and the point $(1, 2)$ to determine the value of b .



3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$

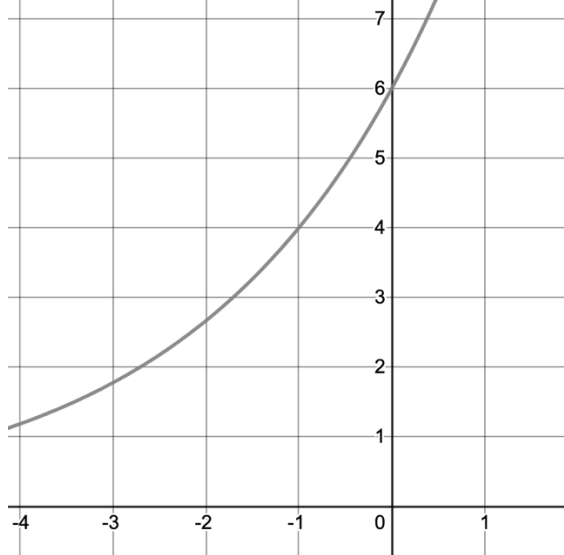
The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	0.4	1.2	3.6	10.8	32.4

4. What is the initial value, a ?
5. How does the value of $f(x)$ change as the x -value increases by 1?
6. Write the function in the form $f(x) = a(b)^x$

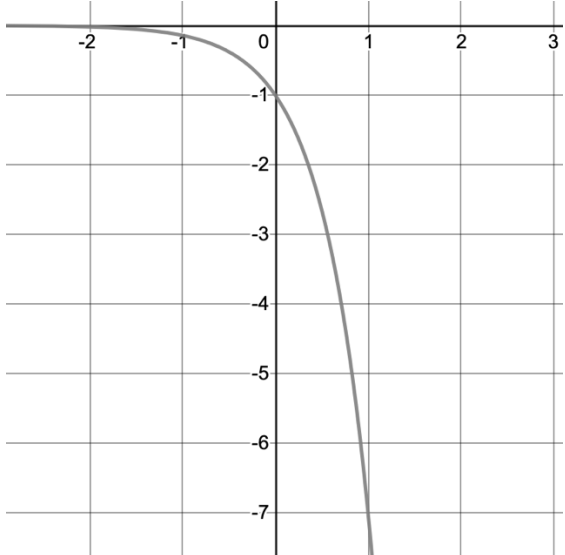
For questions 7-10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

7.



Function

8.



Function

9.

x	y
0	-0.5
1	-4
2	-32
3	-256
4	-2048

Function

10.

x	y
0	3
1	$\frac{15}{4}$
2	$\frac{75}{16}$
3	$\frac{375}{64}$
4	$\frac{1875}{256}$

Function

Algebra 1
Unit 11
Lesson 6 Practice

Name _____
Date _____
Period _____

Use the situation below to answer questions 1-3.

A new MacBook computer costs \$950, but depreciates 10.8% each year as new technology comes out.

1. Is this situation exponential growth or decay?
2. Two students, Quinn and Jyquis were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Quinn	Jyquis
$V(x) = 950(1.108)^x$	$V(x) = 1275(0.892)^x$

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$

Use the situation below to answer questions 4-6.

A new school opened with 134 students in 2021 and plans to increase by 26% per year until they reach full capacity.

4. Is this situation exponential growth or decay?

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.
6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$

7. A three-bedroom house in the suburbs of Miami sold for \$420,625. If the housing prices are expected to increase 3.75% annually, write a function that models the cost of a house, C , x years after it was sold.
8. A small local company reported a profit of \$89,000 last year, but is struggling due to the pandemic. They are reporting a decrease of 12% each year. Write a function that represents the profit, P , after t years.
9. As of 2020, the world population is 9.23 billion people and is expected to grow at a rate of 0.62% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.
10. A culture of 15 bacteria grows at a rate of 250% each hour. Tyler wrote the equation $B(t) = 15(2.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error that Tyler made.